

JARTAI, China

WMO: 535020

Lat: 39.7777N

Lon: 105.7499E

Elev: 1033

StdP: 89.52

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.2	-18.0	-30.7	0.2	-11.9	-28.6	0.3	-11.4	9.0	-3.6	8.0	-2.8	1.8	200	0.528

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	35.8	17.6	34.5	17.0	33.2	16.6	20.8	29.9	19.8	28.5	18.9	27.7	3.6	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.2	14.8	24.2	17.0	13.8	23.5	16.0	12.9	23.1	65.4	29.9	61.6	28.6	58.3	27.7	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.7	8.4	7.3		DB	-23.4	38.8	2.8	1.4	-25.5	39.8	-27.1	40.6	-28.6	41.3	-30.7	42.3
				WB	-23.9	21.7	2.7	1.2	-25.8	22.6	-27.4	23.3	-28.9	23.9	-30.9	24.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.3	-8.6	-3.9	4.2	12.6	18.9	24.1	26.5	24.5	18.8	10.7	1.5	-6.6	(6)
(7)		DBStd	12.81	4.17	4.90	5.15	4.78	3.89	2.97	2.72	2.95	3.56	4.21	4.48	4.24	(7)
(8)		HDD10.0	1999	578	388	189	28	1	0	0	0	43	255	516		(8)
(9)		HDD18.3	3672	836	622	439	178	40	2	0	1	37	239	504	774	(9)
(10)		CDD10.0	2106	0	0	8	107	277	423	512	451	263	64	1	0	(10)
(11)		CDD18.3	738	0	0	0	7	58	174	253	193	51	1	0	0	(11)
(12)		CDH23.3	8617	0	0	3	131	777	2103	3053	2059	468	23	0	0	(12)
(13)		CDH26.7	3947	0	0	0	34	269	989	1593	926	134	2	0	0	(13)
(14)	Wind	WSAvg	3.2	2.7	3.0	3.5	3.8	3.6	3.3	3.2	2.9	2.8	3.1	2.9	(14)	
(15)	Precipitation	PrecAvg	104	1	1	2	5	7	13	25	12	5	2	0	(15)	
(16)		PrecMax	227	6	7	9	21	54	44	94	63	20	20	4	(16)	
(17)		PrecMin	49	0	0	0	0	0	2	6	5	0	0	0	(17)	
(18)		PrecStd	45	2	1	2	6	11	10	22	26	14	5	5	1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.2	12.4	22.2	30.5	33.5	36.6	38.5	36.5	33.4	26.0	16.8	7.4	(19)
MCWB			-0.8	2.9	8.2	12.2	13.8	16.3	19.1	18.3	15.4	11.9	5.8	0.3	(20)	
2%		DB	3.0	9.2	18.6	27.0	31.5	34.9	36.5	34.6	29.8	23.2	14.0	4.5	(21)	
		MCWB	-2.6	1.4	6.2	10.4	13.3	15.6	18.2	17.8	14.7	10.4	4.4	-1.4	(22)	
5%		DB	0.7	7.0	15.9	24.2	29.7	33.4	35.1	33.1	27.8	20.7	11.5	2.3	(23)	
		MCWB	-4.0	0.0	4.9	9.4	12.5	15.3	17.7	17.3	14.1	9.3	3.4	-2.7	(24)	
10%		DB	-1.3	4.8	13.3	21.9	27.8	32.0	33.7	31.6	26.0	18.4	9.1	0.4	(25)	
		MCWB	-5.4	-1.2	3.5	8.4	11.7	15.0	17.5	16.9	13.8	8.4	2.1	-3.9	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.6	4.0	8.8	13.5	16.0	19.4	22.5	18.8	13.6	6.7	0.4	(27)	
MCDB			5.8	9.2	20.1	25.4	26.4	27.6	34.4	30.1	25.5	21.0	14.5	7.0	(28)	
2%		WB	-2.5	1.7	6.7	11.8	14.7	18.2	21.0	20.9	17.4	11.8	5.1	-1.3	(29)	
		MCDB	2.7	8.7	17.2	22.8	27.1	26.7	31.3	29.0	24.6	20.3	12.2	3.9	(30)	
5%		WB	-3.9	0.3	5.4	10.3	13.6	17.4	20.0	19.9	16.3	10.4	3.8	-2.6	(31)	
		MCDB	0.5	6.4	14.7	22.5	26.2	26.6	29.3	28.0	23.5	18.5	10.7	2.0	(32)	
10%		WB	-5.3	-1.0	4.0	9.0	12.6	16.6	19.2	18.8	15.1	9.2	2.5	-3.8	(33)	
		MCDB	-1.4	4.8	12.2	20.9	25.3	26.8	28.3	27.5	22.9	17.1	8.7	0.2	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	13.1	14.2	14.2	13.9	13.6	12.6	11.9	11.7	11.9	12.8	12.1	11.7	(35)
MCDBR			14.8	16.7	17.5	17.4	16.6	15.2	14.3	13.9	14.7	15.7	14.3	13.3	(36)	
MCWBR			10.6	10.6	9.3	8.0	6.5	4.7	4.0	4.0	5.0	7.0	7.9	9.2	(37)	
5% WB		MCDBR	14.6	15.8	15.8	15.1	13.6	11.5	11.5	10.7	10.8	13.6	13.2	12.8	(38)	
		MCWBR	10.5	10.1	8.5	7.1	5.7	3.7	3.5	3.9	4.3	6.7	7.5	8.9	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.329	0.382	0.431	0.478	0.502	0.516	0.499	0.490	0.464	0.411	0.353	0.324	(40)	
(41)		taud	2.102	1.976	1.867	1.795	1.803	1.834	1.949	1.952	1.946	1.997	2.081	2.102	(41)	
(42)		Ebn at Noon	805	807	808	798	788	777	786	780	769	764	766	775	(42)	
(43)		Edh at Noon	111	147	183	210	213	207	183	178	167	142	113	103	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.70	3.71	4.97	6.12	6.87	6.75	6.34	5.88	4.93	4.10	3.01	2.45	(44)	
(45)		RadStd	0.23	0.20	0.31	0.33	0.27	0.47	0.33	0.26	0.35	0.18	0.20	0.17	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	-0.50	N/A	N/A	N/A	N/A	(46)
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+62	N/A	(47)

Nomenclature: See separate page